Churn Model Comparison Demo

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1 IBM Watson Customer Churn Data Set

The `wu_churn` data set is a collection of IBM Watson customer data collected in September 2018. The data contains a factor variable of whether or not the customer has churned, and other related information about the customer.

```
# Install and load the dataset
# install.packages("modeldata")
remove(list = ls())
library(modeldata)
library(tidyverse)
data(wa_churn)
str(wa_churn)
```

```
tibble [7,043 x 20] (S3: tbl_df/tbl/data.frame)
 $ churn            : Factor w/ 2 levels "Yes","No": 2 2 1 2 1 1 2 2 1 2 ...
 $ female           : num [1:7043] 1 0 0 0 1 1 0 1 1 0 ...
```

```
 $ senior_citizen   : int [1:7043] 0 0 0 0 0 0 0 0 0 0 ...
 $ partner          : num [1:7043] 1 0 0 0 0 0 0 0 1 0 ...
 $ dependents       : num [1:7043] 0 0 0 0 0 0 1 0 0 1 ...
 $ tenure           : int [1:7043] 1 34 2 45 2 8 22 10 28 62 ...
 $ phone_service    : num [1:7043] 0 1 1 0 1 1 1 0 1 1 ...
 $ multiple_lines   : Factor w/ 3 levels "No","No phone service",..: 2 1 1 2 1 3 3 2 3 1 ...
 $ internet_service : Factor w/ 3 levels "DSL","Fiber optic",..: 1 1 1 1 2 2 2 1 2 1 ...
 $ online_security  : Factor w/ 3 levels "No","No internet service",..: 1 3 3 3 1 1 1 3 1 3
 $ online_backup    : Factor w/ 3 levels "No","No internet service",..: 3 1 3 1 1 1 3 1 1 3
 $ device_protection: Factor w/ 3 levels "No","No internet service",..: 1 3 1 3 1 3 1 1 3 1
 $ tech_support     : Factor w/ 3 levels "No","No internet service",..: 1 1 1 3 1 1 1 1 3 1
 $ streaming_tv     : Factor w/ 3 levels "No","No internet service",..: 1 1 1 1 1 3 3 1 3 1
 $ streaming_movies : Factor w/ 3 levels "No","No internet service",..: 1 1 1 1 1 3 1 1 3 1
 $ contract         : Factor w/ 3 levels "Month-to-month",..: 1 2 1 2 1 1 1 1 1 2 ...
 $ paperless_billing: num [1:7043] 1 0 1 0 1 1 1 0 1 0 ...
 $ payment_method   : Factor w/ 4 levels "Bank transfer (automatic)",..: 3 4 4 1 3 3 2 4 3 1
 $ monthly_charges  : num [1:7043] 29.9 57 53.9 42.3 70.7 ...
 $ total_charges    : num [1:7043] 29.9 1889.5 108.2 1840.8 151.7 ...
```

2 Report Outcome

The goal of this report is to use the `wu_churn` data set to compare the performance between two models trained to predict whether a customer will churn based on their observed characteristics. Ultimately, this will show quantitative methods of scoring model performance and derive metrics for proving the strongest predictor choice.

3 Data Cleaning

Convert numeric categorical columns to factor

```
# Backup dataset
backup_wa_churn <- wa_churn
factor_cols <- c('female','senior_citizen','partner',
                 'dependents','phone_service','paperless_billing')

# Factoring
wa_churn[factor_cols] <- lapply(
  wa_churn[factor_cols],
  function(x) {factor(x, levels = c(0,1), labels = c('No','Yes'))})
```

```r
# Comparison check
head(wa_churn,3)
```

```
# A tibble: 3 x 20
  churn female senior_citizen partner dependents tenure phone_service
  <fct> <fct>  <fct>          <fct>   <fct>       <int> <fct>
1 No    Yes    No             Yes     No              1 No
2 No    No     No             No      No             34 Yes
3 Yes   No     No             No      No              2 Yes
# i 13 more variables: multiple_lines <fct>, internet_service <fct>,
#   online_security <fct>, online_backup <fct>, device_protection <fct>,
#   tech_support <fct>, streaming_tv <fct>, streaming_movies <fct>,
#   contract <fct>, paperless_billing <fct>, payment_method <fct>,
#   monthly_charges <dbl>, total_charges <dbl>
```

```r
head(backup_wa_churn,3)
```

```
# A tibble: 3 x 20
  churn female senior_citizen partner dependents tenure phone_service
  <fct>  <dbl>          <int>   <dbl>      <dbl>  <int>         <dbl>
1 No         1              0       1          0      1             0
2 No         0              0       0          0     34             1
3 Yes        0              0       0          0      2             1
# i 13 more variables: multiple_lines <fct>, internet_service <fct>,
#   online_security <fct>, online_backup <fct>, device_protection <fct>,
#   tech_support <fct>, streaming_tv <fct>, streaming_movies <fct>,
#   contract <fct>, paperless_billing <dbl>, payment_method <fct>,
#   monthly_charges <dbl>, total_charges <dbl>
```

Inspect multicollinearity in the continuous subset. As expected, total charges is highly correlated with monthly charges and tenure and can be dropped to avoid model redundancy.

```r
# Continous subset
continuous_cols <- c('tenure','monthly_charges','total_charges')

# Pearson correlation matrix
cor(na.omit(wa_churn[continuous_cols]))
```

```
                tenure monthly_charges total_charges
tenure       1.0000000       0.2468618     0.8258805
```

```
monthly_charges 0.2468618       1.0000000     0.6510648
total_charges   0.8258805       0.6510648     1.0000000
```

```
# Drop redundant total_charges
wa_churn <- wa_churn |>
  select(-total_charges)
```

4 Data Inspection

Understanding the value possibilities for each variable and finding relationships between them will be helpful in determining which should be included in the model. Internet amenities are influenced by the existence of internet service. This will have to be considered when designing the model.

```
# Columns
unique_cols <- c('multiple_lines', 'internet_service', 'online_security',
                 'online_backup', 'device_protection', 'tech_support',
                 'streaming_tv', 'streaming_movies', 'contract')

# Unique value analysis
unique_list <- as.data.frame(lapply(wa_churn[unique_cols], unique))
unique_list
```

```
    multiple_lines internet_service     online_security       online_backup
1 No phone service              DSL                  No                 Yes
2               No      Fiber optic                 Yes                  No
3              Yes               No No internet service No internet service
    device_protection        tech_support        streaming_tv
1                  No                  No                  No
2                 Yes                 Yes                 Yes
3 No internet service No internet service No internet service
     streaming_movies       contract
1                  No Month-to-month
2                 Yes       One year
3 No internet service       Two year
```

5 Model Creation

Two models will be created for this report:

- The `claude_model` will represent the prediction capabilities of a general LLM. It will be trained on two major predictor variables: `tenure` and `paperless_billing`
- The `athlera_model` will represent the prediction capabilities of Athlera's proprietary deterministic, guard-railed LLM. It will be trained on the following predictor variables:
  - `female` - gender of customer
  - `senior_citizen` - age bracket
  - `tenure` - length of subscription
  - `phone_service` - whether they are subscribed to a phone service
  - `contract` - billing frequency
  - `paperless_billing` - method of payment
  - `monthly_charges` - monthly payment for services

The